

$$\sigma, \tau ::= \dots \mid \sigma \times \tau$$

$$M, N ::= \dots \mid \langle M, N \rangle \mid \pi_1(M) \mid \pi_2(M)$$

Definir como macro la función $\text{curry}_{\sigma, \tau, \delta}$ que sirve para currificar funciones que reciben pares como argumento.

$$\text{curry}_{\delta, \tau, \delta} = \lambda f: \delta \times \tau \rightarrow \delta. \lambda x: \delta. \lambda y: \tau. f \langle x, y \rangle$$

$$\frac{\Gamma \vdash M: \delta \quad \Gamma \vdash N: \tau}{\Gamma \vdash \langle M, N \rangle: \delta \times \tau} \text{T-PAR} \qquad \frac{\Gamma \vdash M: \delta \times \tau}{\Gamma \vdash \pi_1(M): \delta} \text{T-}\pi_1$$

$$\frac{\Gamma \vdash M: \delta \times \tau}{\Gamma \vdash \pi_2(M): \tau} \text{T-}\pi_2$$

$$\pi_1(\text{false}) \quad \times$$

$$\pi_1(\langle \text{true}, \text{false} \rangle) \rightarrow \underline{\quad}$$

$$M ::= \dots \mid \langle M, M \rangle \mid \pi_1(M) \mid \pi_2(M)$$

$$V, W ::= \text{true} \mid \text{false} \mid \lambda x: \delta. M \mid \text{zero} \mid \text{succ}(V) \mid \langle V, W \rangle$$

$$\pi_1(\langle \text{true}, \text{false} \rangle) \rightarrow \text{true}$$

* Determinismo

$$\left. \begin{array}{l} M \rightarrow N_1 \\ M \rightarrow N_2 \end{array} \right\} \Rightarrow N_1 = N_2$$

* Preservación de tipo

$$\left. \begin{array}{l} \Gamma \vdash M: \delta \\ M \rightarrow N \end{array} \right\} \Rightarrow \Gamma \vdash N: \delta$$

* Progreso

$$\emptyset \vdash M: \delta \Rightarrow \left\{ \begin{array}{l} \text{o bien } M \text{ es un valor} \\ \text{o bien } \exists N \text{ t.q. } M \rightarrow N \end{array} \right.$$

* Terminación

$$\emptyset \vdash M: \delta \Rightarrow \underline{\text{no pasa}} \text{ que } M \rightarrow N \rightarrow \dots \quad \text{cadena } \infty$$

$$\pi_1(\langle \quad, \quad \rangle)$$

$$M ::= \dots \mid \langle M, M \rangle \mid \pi_1(M) \mid \pi_2(M)$$

$$V, W ::= \text{true} \mid \text{false} \mid \lambda x:\delta. M \mid \text{zero} \mid \text{succ}(V) \mid \langle V, W \rangle$$

$$\overline{M \rightarrow M'} \quad \text{M no es un valor}$$

$$\frac{M \rightarrow M'}{\pi_1(M) \rightarrow \pi_1(M')} \quad \text{E-}\pi_1$$

$$\frac{}{\pi_1(\langle V_1, V_2 \rangle) \rightarrow V_1} \quad \text{E-}\pi_1 \text{ PAR}$$

$$\pi_1((\lambda x:\text{Bool} \times \text{Bool}. x) \langle \text{true}, \text{false} \rangle)$$

$$\longrightarrow \pi_1(\langle \text{true}, \text{false} \rangle)$$

$$\begin{array}{c} \text{E-}\pi_1 \\ + \\ \beta \end{array}$$

$$\emptyset \vdash \langle \langle \text{true}, \text{false} \rangle, \text{zero} \rangle : (\text{Bool} \times \text{Bool}) \times \text{Nat}$$

$$\frac{M \rightarrow M'}{\pi_2(M) \rightarrow \pi_2(M')} \quad \text{E-}\pi_2$$

$$\frac{}{\pi_2(\langle V_1, V_2 \rangle) \rightarrow V_2} \quad \text{E-}\pi_2 \text{ PAR}$$

$$\frac{M \rightarrow M'}{\langle M, N \rangle \rightarrow \langle M', N \rangle} \quad \text{E-PAR1}$$

$$\frac{N \rightarrow N'}{\langle V, N \rangle \rightarrow \langle V, N' \rangle} \quad \text{E-PAR2}$$

¿Qué problema introduce agregar la siguiente regla?

$$\pi_1(\langle M, N \rangle) \rightarrow M$$

$$\pi_1((\lambda x:\text{Bool} \times \text{Nat}. x) \langle \text{true}, 0 \rangle)$$

$$\frac{M \rightarrow M'}{\pi_1(M) \rightarrow \pi_1(M')} \quad \text{E-}\pi_1$$

$$\frac{}{\pi_1(\langle V_1, V_2 \rangle) \rightarrow V_1} \quad \text{E-}\pi_1 \text{ PAR}$$

$$\frac{M \rightarrow M'}{\pi_2(M) \rightarrow \pi_2(M')} \quad \text{E-}\pi_2$$

$$\frac{}{\pi_2(\langle V_1, V_2 \rangle) \rightarrow V_2} \quad \text{E-}\pi_2 \text{ PAR}$$

$$\frac{M \rightarrow M'}{\langle M, N \rangle \rightarrow \langle M', N \rangle} \quad \text{E-PAR1}$$

$$\frac{N \rightarrow N'}{\langle V, N \rangle \rightarrow \langle V, N' \rangle} \quad \text{E-PAR2}$$

$$\pi_1(\langle (\lambda x:\text{Bool}. x) \text{true}, \text{false} \rangle) \xrightarrow{\text{MAL}} (\lambda x:\text{Bool}. x) \text{true} \quad \times$$

$$\searrow \pi_1(\langle \text{true}, \text{false} \rangle)$$

$$\begin{array}{c}
\frac{}{x:\text{Nat} \vdash x:\text{Nat}} \text{T-VAR} \quad \frac{}{x:\text{Nat} \vdash \text{true}:\text{Bool}} \text{T-TRUE} \\
\frac{}{x:\text{Nat} \vdash \langle x, \text{true} \rangle : \text{Nat} \times \text{Bool}} \text{T-PAR} \\
\frac{}{\emptyset \vdash \lambda x:\text{Nat}. \langle x, \text{true} \rangle : \text{Nat} \rightarrow \text{Nat} \times \text{Bool}} \text{T-ABS} \quad \frac{}{\emptyset \vdash 0 : \text{Nat}} \text{T-ZERO} \\
\frac{}{\emptyset \vdash (\lambda x:\text{Nat}. \langle x, \text{true} \rangle) 0 : \text{Nat} \times \text{Bool}} \text{T-APP} \\
\frac{}{\emptyset \vdash \pi_1((\lambda x:\text{Nat}. \langle x, \text{True} \rangle) 0) : \text{Nat}} \text{T-}\pi_1
\end{array}$$

$$\begin{array}{c}
\frac{\Gamma \vdash \delta \quad \Gamma \vdash \tau}{\Gamma \vdash \delta \wedge \tau} \text{T-PAR} \quad \frac{\Gamma \vdash \delta \wedge \tau}{\Gamma \vdash \delta} \text{T-}\pi_1 \\
\frac{\Gamma \vdash \delta \wedge \tau}{\Gamma \vdash \tau} \text{T-}\pi_2
\end{array}$$

$$\pi_1((\lambda x:\text{Nat}. \langle x, \text{True} \rangle) 0) \xrightarrow[\beta]{E-\pi_1} \pi_1(\langle 0, \text{true} \rangle) \xrightarrow{E-\pi_1 \text{ PAR}} 0$$

$$\pi_1(\pi_1(\lambda x:\text{Nat}, \langle \langle x, x \rangle, \text{true} \rangle) 0) \xrightarrow[\beta]{E-\pi_1, E-\pi_1} \pi_1(\pi_1(\langle \langle 0, 0 \rangle, \text{true} \rangle))$$

$$\begin{array}{c}
\frac{M \rightarrow M'}{\pi_1(M) \rightarrow \pi_1(M')} E-\pi_1 \quad \frac{}{\pi_1(\langle V_1, V_2 \rangle) \rightarrow V_1} E-\pi_1 \text{ PAR} \\
\frac{M \rightarrow M'}{\pi_2(M) \rightarrow \pi_2(M')} E-\pi_2 \quad \frac{}{\pi_2(\langle V_1, V_2 \rangle) \rightarrow V_2} E-\pi_2 \text{ PAR} \\
\frac{M \rightarrow M'}{\langle M, N \rangle \rightarrow \langle M', N \rangle} E-\text{PAR1} \quad \frac{N \rightarrow N'}{\langle V, N \rangle \rightarrow \langle V, N' \rangle} E-\text{PAR2}
\end{array}$$

$$\begin{array}{l}
\sigma ::= \dots \mid \sigma + \sigma = \sigma, \tau ::= \dots \mid \delta + \tau \\
M ::= \dots \mid \text{left}_{\sigma}(M) \mid \text{right}_{\sigma}(M) \mid \\
\text{case } M \text{ of left}(x) \rightsquigarrow M \parallel \text{right}(y) \rightsquigarrow M
\end{array}$$

$$\frac{\Gamma \vdash M : \tau}{\Gamma \vdash \text{left}_\delta(M) : \tau + \delta} \text{ T-LEFT}$$

$$\vdash \text{left}(\text{zero}) : \text{Nat} + \text{O} ?$$

$$\vdash \text{left}_{\text{Bool}}(\text{zero}) : \text{Nat} + \text{Bool}$$

$$\frac{\Gamma \vdash M : \tau}{\Gamma \vdash \text{right}_\delta(M) : \delta + \tau} \text{ T-RIGHT}$$

$$\vdash \text{left}_{\text{Nat} \rightarrow \text{Bool}}(\text{zero}) : \text{Nat} + (\text{Nat} \rightarrow \text{Bool})$$

$$\frac{\Gamma, x : \delta \vdash M : \tau}{\Gamma \vdash \lambda x : \delta. M : \delta \rightarrow \tau} \text{ T-ABS}$$

$$\vdash \text{zero} : \text{Nat}$$

$$\frac{\Gamma \vdash M : \delta + \tau \quad \Gamma, x : \delta \vdash N : \rho \quad \Gamma, y : \tau \vdash O : \rho}{\Gamma \vdash \text{case } M \text{ of } \text{left}(x) \rightsquigarrow N \mid \text{right}(y) \rightsquigarrow O : \rho} \text{ T-CASE}$$

$$\bullet V ::= \dots \mid \text{left}_\delta(V) \mid \text{right}_\delta(V)$$

true

$\text{left}_{\text{Bool}}(\text{zero})$

$\text{left}_{\text{Bool}}(\text{succ}(\text{zero}))$

$\text{left}_{\text{Bool}}(\text{if true then } \underline{0} \text{ else } \underline{0})$

$$\vdash \text{left}_{\text{Bool}}(\underline{0}) : \text{Nat} + \text{Bool}$$

$$\vdash \underline{0} : \text{Nat}$$

$$\frac{M \rightarrow M'}{\text{left}_\delta(M) \rightarrow \text{left}_\delta(M')} \text{ E-LEFT}$$

$$\frac{M \rightarrow M'}{\text{right}_\delta(M) \rightarrow \text{right}_\delta(M')} \text{ E-RIGHT}$$

$$\frac{M \rightarrow M'}{\text{case } M \text{ of } \text{left}(x) \rightsquigarrow N \mid \text{right}(y) \rightsquigarrow O \rightarrow \text{case } M' \text{ of } \text{left}(x) \rightsquigarrow N \mid \text{right}(y) \rightsquigarrow O} \text{ E-CASE}$$

case M of left(x) $\rightsquigarrow$ N $\mid$ right(y) $\rightsquigarrow$ O

$\rightarrow$

case M' of left(x) $\rightsquigarrow$ N $\mid$ right(y) $\rightsquigarrow$ O

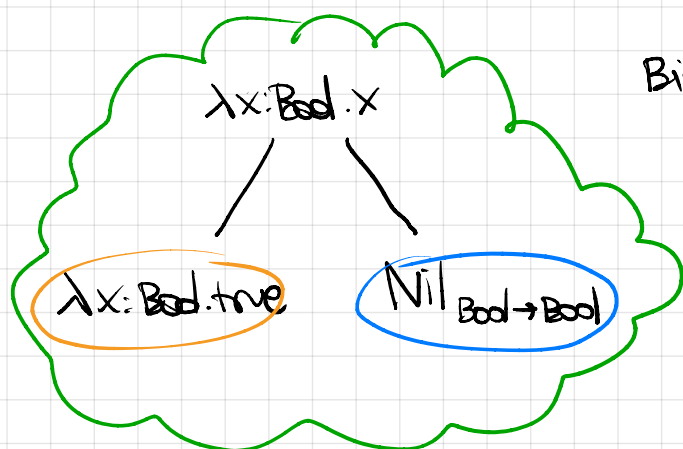
case $\text{left}_\delta(V)$ of $\text{left}(x) \rightsquigarrow N \parallel \text{right}(y) \rightsquigarrow O$
 $\longrightarrow N \{x := V\}$

case $\text{right}_\delta(V)$ of $\text{left}(x) \rightsquigarrow N \parallel \text{right}(y) \rightsquigarrow O$
 $\longrightarrow O \{y := V\}$

$\frac{\Gamma \vdash M : \delta + \tau \quad \Gamma, x : \delta \vdash N : \rho \quad \Gamma, y : \tau \vdash O : \rho}{\Gamma \vdash \text{case } M \text{ of } \text{left}(x) \rightsquigarrow N \parallel \text{right}(y) \rightsquigarrow O : \rho}$ T-CASE

$\sigma ::= \dots \mid AB_\sigma$

$M, N, O ::= \dots \mid \text{Nil}_\sigma \mid \text{Bin}(M, N, O) \mid \text{raíz}(M) \mid \text{der}(M) \mid$
 $\text{izq}(M) \mid \text{esNil}(M)$



$\text{Bin}(\text{Bin}(\text{Nil}_{B \rightarrow B}, \lambda x : \text{Bool}. \text{true}, \text{Nil}_{B \rightarrow B}),$
 $\lambda x : \text{Bool}. x, \text{Nil}_{\text{Bool} \rightarrow \text{Bool}})$

$\frac{}{\Gamma \vdash \text{Nil}_\delta : AB_\delta}$ T-NIL

$\frac{\Gamma \vdash M : AB_\delta \quad \Gamma \vdash N : \delta \quad \Gamma \vdash O : AB_\delta}{\Gamma \vdash \text{Bin}(M, N, O) : AB_\delta}$ T-BIN

$\text{Nil}_{\text{Bool}} \neq \text{Nil}_{\text{Nat}}$

$\frac{\Gamma \vdash M : AB_\delta}{\Gamma \vdash \text{raíz}(M) : \delta}$ T-RAÍZ

$\frac{\Gamma \vdash M : AB_\delta}{\Gamma \vdash \text{izq}(M) : AB_\delta}$ T-IZQ

$\frac{\Gamma \vdash M : AB_\delta}{\Gamma \vdash \text{esNil}(M) : \text{Bool}}$ T-ESNIL

$\frac{\Gamma \vdash M : AB_\delta}{\Gamma \vdash \text{der}(M) : AB_\delta}$ T-DER

$\sigma ::= \dots \mid AB_\sigma$

$M, N, O ::= \dots \mid Nil_\sigma \mid Bin(M, N, O) \mid raíz(M) \mid der(M) \mid izq(M) \mid esNil(M)$

$V ::= \dots \mid Nil_\delta \mid Bin(V, V, V)$

$$\frac{M \rightarrow M'}{Bin(M, N, O) \rightarrow Bin(M', N, O)} \text{E-BIN1}$$

$$\frac{N \rightarrow N'}{Bin(V, N, O) \rightarrow Bin(V, N', O)} \text{E-BIN2}$$

$$\frac{O \rightarrow O'}{Bin(V_1, V_2, O) \rightarrow Bin(V_1, V_2, O')} \text{E-BIN3}$$

$$\frac{M \rightarrow M'}{raíz(M) \rightarrow raíz(M')} \text{E-RAÍZ}$$

$$\frac{}{raíz(Bin(V_1, V_2, V_3)) \rightarrow V_2} \text{E-RAÍZBIN}$$

COMENTARIO

$raíz(Nil_\delta) \rightarrow$ ~~$\rightarrow$~~

$izq(Nil_\delta) \rightarrow$ ~~$\rightarrow$~~

$der(Nil_\delta) \rightarrow$ ~~$\rightarrow$~~

pendientes
progreso.

$$\frac{M \rightarrow M'}{izq(M) \rightarrow izq(M')} \text{E-IZQ}$$

$$\frac{M \rightarrow M'}{der(M) \rightarrow der(M')} \text{E-DER}$$

$$\frac{}{izq(Bin(V_1, V_2, V_3)) \rightarrow V_1} \text{E-IZQBIN}$$

$$\frac{}{der(Bin(V_1, V_2, V_3)) \rightarrow V_3} \text{E-DERBIN}$$

$$\frac{M \rightarrow M'}{esNil(M) \rightarrow esNil(M')} \text{E-ESNIL}$$

$$\frac{}{esNil(Bin(V_1, V_2, V_3)) \rightarrow \text{false}} \text{E-ESNILBIN}$$

$$\frac{}{esNil(Nil_\delta) \rightarrow \text{true}} \text{E-ESNILNIL}$$

$\delta ::= \dots \mid AB_\delta$

$M, N, O ::= \dots \mid Nil_\sigma \mid Bin(\overline{M}, \overline{N}, \overline{O}) \mid$
 $case \overline{M} \text{ of } Nil \rightsquigarrow N; Bin(i, r, d) \rightsquigarrow \overline{O}$

$$\frac{}{\Gamma \vdash Nil_\delta : AB_\delta} T-NIL \quad \frac{\Gamma \vdash M : AB_\delta \quad \Gamma \vdash N : \delta \quad \Gamma \vdash O : AB_\delta}{\Gamma \vdash Bin(M, N, O) : AB_\delta} T-BIN$$

$$\frac{\Gamma \vdash M : AB_\varepsilon \quad \Gamma \vdash N : \delta \quad \Gamma, i : AB_\varepsilon, r : \varepsilon, d : AB_\varepsilon \vdash O : \delta}{\Gamma \vdash case M \text{ of } Nil \rightsquigarrow N; Bin(i, r, d) \rightsquigarrow O : \delta} T-CASEAB$$

• $V ::= \dots \mid Nil_\delta \mid Bin(V, V, V) \quad \checkmark$

•

$$\frac{M \rightarrow M'}{Bin(M, N, O) \rightarrow Bin(M', N, O)} E-BIN1 \quad \frac{N \rightarrow N'}{Bin(V, N, O) \rightarrow Bin(V, N', O)} E-BIN2$$

$$\frac{O \rightarrow O'}{Bin(V_1, V_2, O) \rightarrow Bin(V_1, V_2, O')} E-BIN3$$

$$\frac{M \rightarrow M'}{case M \text{ of } Nil \rightsquigarrow N; Bin(i, r, d) \rightsquigarrow O \rightarrow case M' \text{ of } Nil \rightsquigarrow N; Bin(i, r, d) \rightsquigarrow O} E-CASEAB$$

$$\frac{}{case Nil_\delta \text{ of } Nil \rightsquigarrow N; Bin(i, r, d) \rightsquigarrow O \rightarrow N} E-CASENIL$$

$$\frac{}{case Bin(V_1, V_2, V_3) \text{ of } Nil \rightsquigarrow N; Bin(i, r, d) \rightsquigarrow O \rightarrow O \{i := V_1, r := V_2, d := V_3\}} E-CASEBIN$$

case if $(\lambda x : \text{Bool}.x) \text{ True then Bin}(\text{Nil}_{\text{Nat}}, \underline{1}, \text{Nil}_{\text{Nat}}) \text{ else Nil}_{\text{Nat}}$
of $\text{Nil} \rightsquigarrow \text{False}$; $\text{Bin}(i, r, d) \rightsquigarrow \text{iszero}(r)$

$\longrightarrow$ case if true then $\text{Bin}(\text{Nil}_{\text{Nat}}, \underline{1}, \text{Nil}_{\text{Nat}})$
E-CASEAB
E-IF
 β else Nil_{Nat} of
 $\text{Nil} \rightsquigarrow \text{False}$;
 $\text{Bin}(i, r, d) \rightsquigarrow \text{iszero}(r)$

$\longrightarrow$ case $\text{Bin}(\text{Nil}_{\text{Nat}}, \underline{1}, \text{Nil}_{\text{Nat}})$ of
E-CASEAB
E-IFTRUE
 $\text{Nil} \rightsquigarrow \text{False}$;
 $\text{Bin}(i, r, d) \rightsquigarrow \text{iszero}(r)$

E-CASEBIN
 $\longrightarrow$ $\text{iszero}(r) \{ i := \text{Nil}_{\text{Nat}}, r := \underline{1}, d := \text{Nil}_{\text{Nat}} \} =$
 $\text{iszero}(\underline{1}) = \text{iszero}(\text{succ}(\text{zero}))$

$\longrightarrow$ false
E-ISZERO

$\frac{M \rightarrow M'}{\text{if } M \text{ then } N \text{ else } O} \text{E-IF}$

$\longrightarrow$
if M' then N
else O

$\frac{}{\text{if true then } N \text{ else } O \rightarrow N} \text{E-IFTRUE}$

$\frac{}{\text{if false then } N \text{ else } O \rightarrow O} \text{E-IFFALSE}$

$M, N, O ::= \dots \mid \text{Nil}_{\sigma} \mid \text{Bin}(M, N, O) \mid$
case M of $\text{Nil} \rightsquigarrow N$; $\text{Bin}(i, r, d) \rightsquigarrow O$

$\text{eval}_{\delta} = \lambda x : \text{AB}_{\delta}. \text{case } x \text{ of Nil} \rightsquigarrow \text{true} ; \text{Bin}(i, r, d) \rightsquigarrow \text{false}$

$\text{raiz}_{\delta} = \lambda x : \text{AB}_{\delta}. \text{case } x \text{ of Nil} \rightsquigarrow \underline{1}_{\delta} ; \text{Bin}(i, r, d) \rightsquigarrow r$
usa Fix (Punto Fijo, 2da teórica)

$\text{izq}_{\delta} = \lambda x : \text{AB}_{\delta}. \text{case } x \text{ of Nil} \rightsquigarrow \underline{1}_{\text{AB}_{\delta}} ; \text{Bin}(i, r, d) \rightsquigarrow i$

Definir una nueva extensión que incorpore expresiones de la forma $\text{map}(M, N)$, donde N es un árbol y M una función que se aplicará a cada uno de los elementos de N .

$M, N ::= \dots \mid \text{map}(M, N)$

• $\frac{\Gamma \vdash M : \delta \rightarrow \rho \quad \Gamma \vdash N : \text{AB}_{\delta}}{\Gamma \vdash \text{map}(M, N) : \text{AB}_{\rho}} \text{T-MAP}$

- $V ::= \dots$ | no λ expression values.

$$\frac{M \rightarrow M'}{\text{map}(M, N) \rightarrow \text{map}(M', N)} \quad \text{E-MAP1}$$

$$\frac{N \rightarrow N'}{\text{map}(V, N) \rightarrow \text{map}(V, N')} \quad \text{E-MAP2}$$

$$\frac{\vdash V: \tau \rightarrow \rho}{\text{map}(V, \text{Nil}_\tau) \rightarrow \text{Nil}_\rho} \quad \text{E-MAPNIL}$$

$\text{map}(\lambda x: \text{Bool}. \text{zero}, \text{Nil}_{\text{Bool}}) \rightarrow \text{Nil}_{\text{Nat}}$ E-MAPNIL
 $\vdash \lambda x: \text{Bool}. \text{zero}: \text{Bool} \rightarrow \text{Nat}$

$$\frac{}{\text{map}(V, \text{Bin}(V_1, V_2, V_3)) \rightarrow \text{Bin}(\text{map}(V, V_1), \text{map}(V, V_2), \text{map}(V, V_3))} \quad \text{E-MAPBIN}$$